Curriculum for BI3016 Molecular cell biology autumn 2026

Textbook: Molecular biology of the cell, 7th edition, Alberts et al. (2022)

Lecture 1

Chapter 7:	Control of Gene Expression
Part 1:	An overview of gene control (pages 397-410)
	Transcription regulators switch genes on and off (410-423)
	The molecular mechanism that create specialized cell types (423-435)

Lecture 2

Chapter 7:	Control of Gene Expression
Part 2:	Mechanisms that reinforce cell memory in plants and animals (435-445).
	Post transcriptional controls (445-462).
	Regulation of gene expression by non-coding RNAs (462-471).

Lecture 3.

Chapter 15:	Cell Signalling
Part 1:	Principles of cell signalling (873-892).

Lecture 4.

Chapter 15:	Cell Signalling
Part 2:	Signalling through G-protein-coupled receptors (892-910).

Lecture 5.

Chapter 15	Cell Signalling
Part 3:	Signalling through enzyme-coupled cell surface receptors (911-927). Alternative signalling routes in gene regulation (928-940)

Lecture 6.

Chapter 15:	Plant signalling mechanisms
	Signalling in plants (940-945) Alberts et al. 2022 Articles: Will be published later

Lecture 7.

Chapter 16:	The Cytoskeleton
	Actin (957-975).
	Myosin and actin (976-986).
	Microtubules (987-1007).
	Cell polarity and coordination of the cytoskeleton (1016-1023)

Lecture 8.

Chapter 17:	The Cell Cycle
	Overview of the cell cycle (1027-1031)
	The cell-cycle control system (1031-1042)
	S-phase and Mitosis (1042-1064)

	Cytokinesis (1064-1070)
	Control of cell division and cell growth (1077-1084)
Lecture 9.	
Chapter 18:	Cell death
	Apoptosis (1089-1102)
	Leading edge review article:
	Cell death: Newton K. et al. Cell 187, January 18, 2024 (pages 235-248).
Lecture 10	
Chapter 19:	Cells in their social context: Cell junctions and the extracellular matrix
	Cell-cell junctions (1105-1127)
	The extracellular matrix of animals (1141-1146)
	Cell-matrix junctions (1147-1154)
Lecture 11.	
Chapter 20:	Cancer
	Cancer as a microevolutionary process (1163-1178)
	Cancer-critical genes: How they are found and what they do (1178-1198)
	Cancer prevention and treatment: present and future (1198-1213)
Lecture 12.	
Chapter 21:	Development of multicellular organisms
	Overview of development (1217-1226)
	Mechanisms of pattern formation (1226-1248)